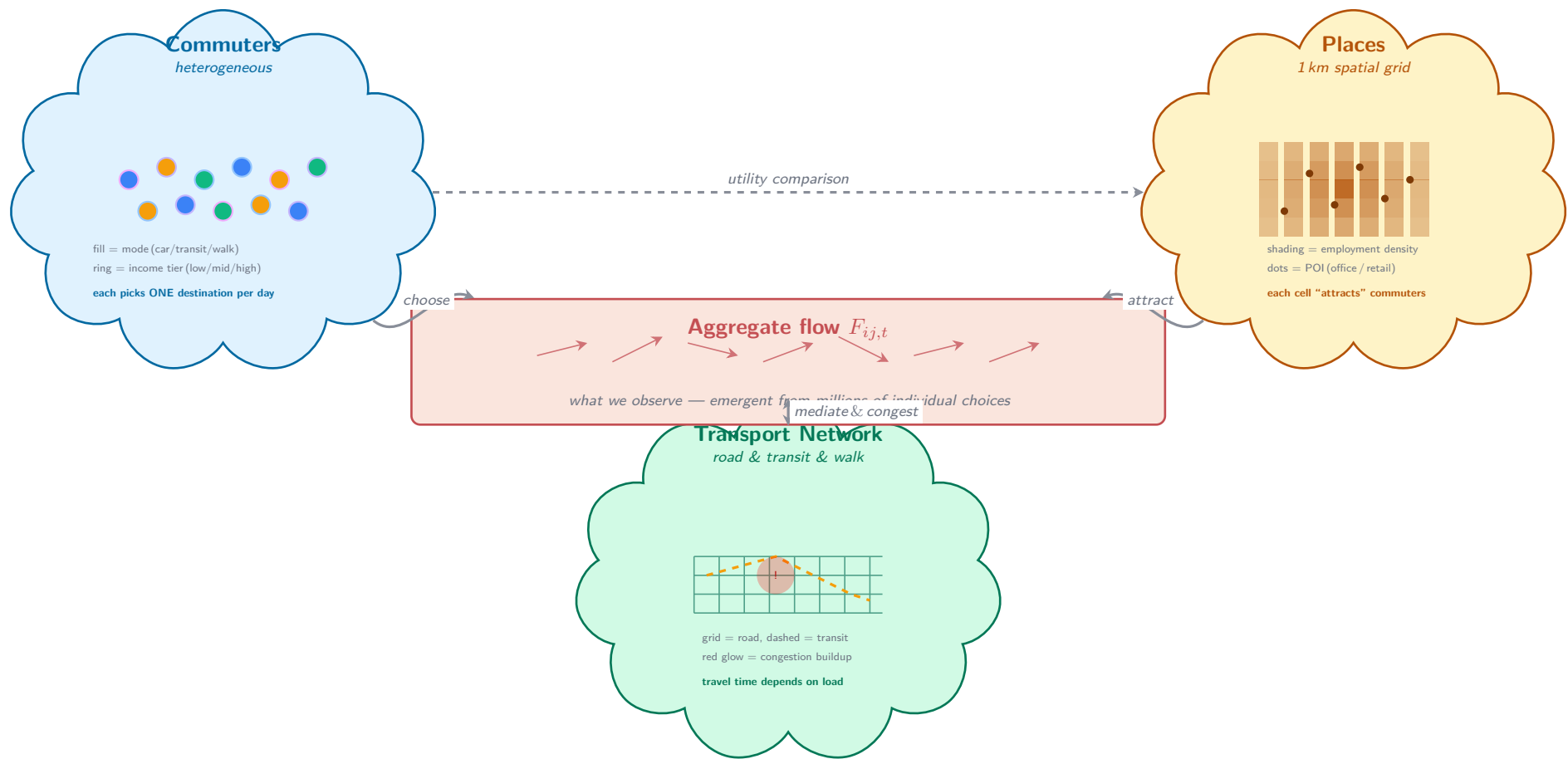


Conceptual Framework — three interacting entities, an observable flow

commuting as an emergent phenomenon, before any modeling apparatus



Inverse problem: from observed flow $F_{ij,t}$, recover the underlying trade-offs

QUANTIFY

Forward use: given the same trade-offs, predict flow under any city configuration

What this lens lets us see — the trade-offs commuters implicitly make



Once these trade-offs are visible, we can ask: “if the city changed (more jobs here, flexible hours there), would the same trade-offs still hold? and how would the flow redistribute?”

The conceptual framework above is data-and-question-side only. The modeling apparatus that makes the leap from observed $F_{ij,t}$ to recovered trade-offs (and back, under counterfactual) lives in Figure 1 (training) and Figure 2 (forward simulation).